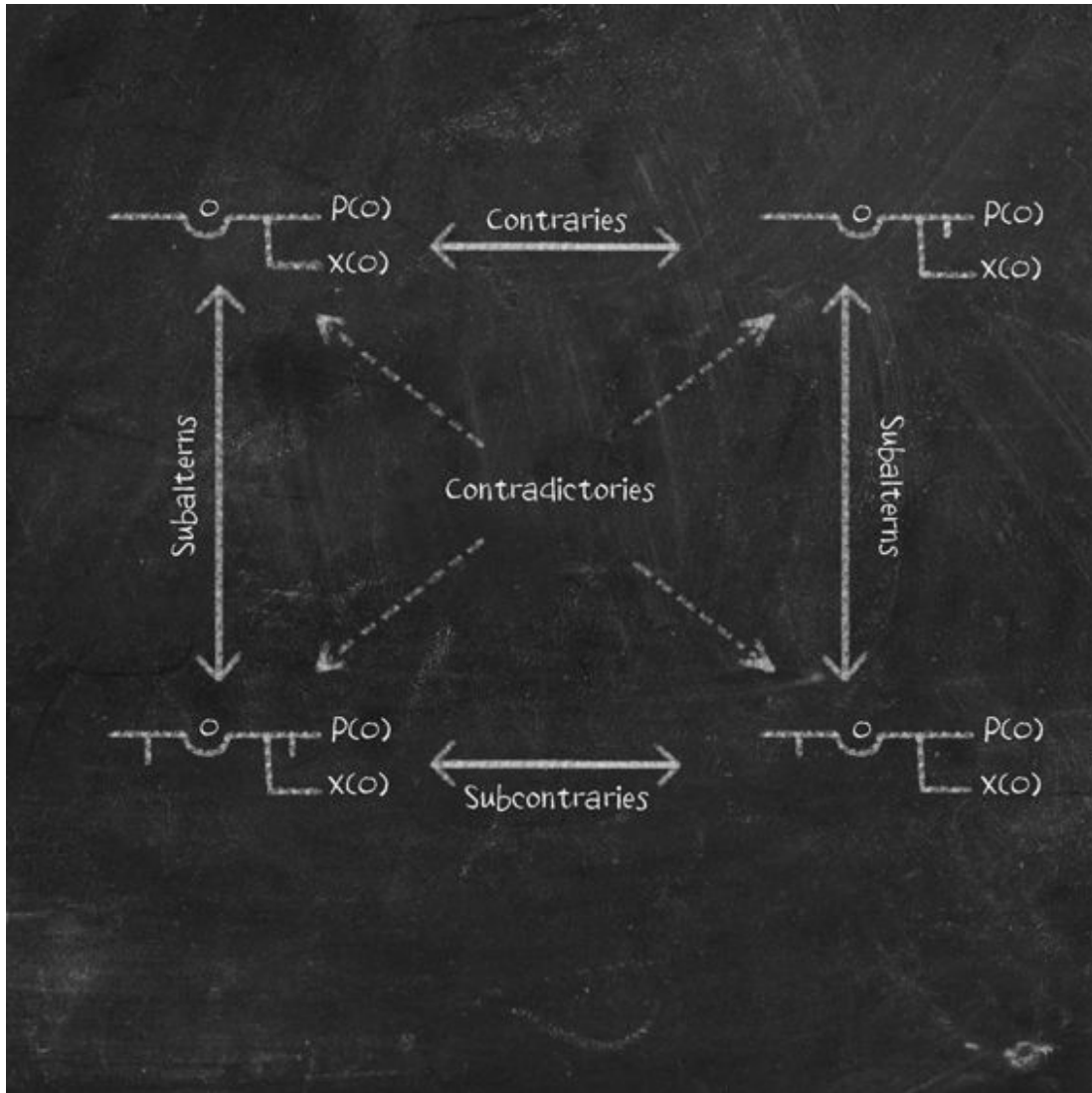


Mathematics and logic

In his book *Begriffsschrift* (1879), German mathematician Gottlob Frege overturned views of logic that had held firm for more than 2,000 years. At the time, it was regarded as a set of 'rules' derived from the way that we think. But Frege proposed, for the first time, that logic is objective and has nothing to do with the way we use it. Logical propositions are objective truths, and either follow one from another, or do not, irrespective of human psychology. Frege also pointed out that mathematics consists of a series of arguments and demonstrations to show how one thing follows from another, and therefore has the same principles as logic. Until this point, the prevailing view had been that mathematics, too, was a human creation, like language, but Frege's arguments showed that mathematics was objective and universal. We do not create it, but discover it in the same way that we discover physical laws. Frege's linking of mathematics and logic also showed that there was far more to logic than the syllogism, making it a more powerful tool for philosophy.



In his *Begriffsschrift*, Frege developed a complex system of conceptual notation for analysing philosophical problems. Here, his notations applied to the standard arguments of the syllogism (see [Logic and the syllogism](#)).